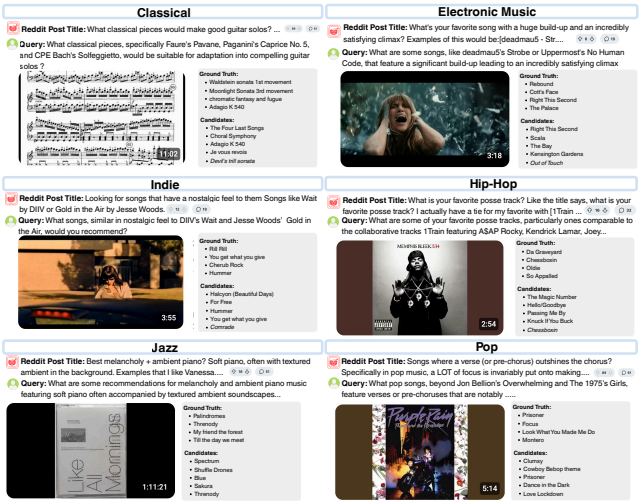

Fig. 2: Representative examples from MusiCRS showing Reddit conversations, derived queries, audio, and candidates across genres.

- We introduce **MusiCRS**, the first benchmark for audio-centric conversational music recommendation, built from 477 Reddit discussions across seven genres with comprehensive audio grounding via validated YouTube links.
- We establish rigorous evaluation protocols across three input modalities and provide comprehensive benchmarking of audio-LLMs, retrieval models, and traditional approaches, revealing substantial performance gaps in current systems.
- We uncover a fundamental limitation in current multimodal models: optimal performance frequently occurs in single-modality settings, showing that models fail to ground abstract musical concepts in audio content.

2. MUSICRS DATASET

2.1. Dataset Construction

Large-Scale Reddit Mining and Filtering. We collected 2.7 million Reddit submissions and 28.5 million comments from high-activity, music-oriented subreddits, including *r/classicalmusic*, *r/hiphopheads*, *r/electronicmusic*, *r/metal*, *r/popheads*, *r/indieheads*, and *r/jazz*, following prior large-scale Reddit-based efforts in conversational AI [16, 17]. We applied a three-stage filtering pipeline to ensure both musical grounding and conversational quality. First, only posts with valid YouTube links were retained, yielding $46,218$ threads linked to external music resources. Second, threads with fewer than three first-level replies or comments under five characters were removed, leaving $10,167$ multi-turn discussions with substantive context [10]. Finally, we discarded low-interaction threads during human annotation, ensuring the final dataset contained coherent and contextually rich exchanges.

Music Entity Extraction and Query Generation. We used LLMs to transform Reddit discussions into structured inputs for downstream tasks. Entity extraction was performed with `Qwen2.5-7B` [18], identifying and categorizing songs, artists, and albums from post titles, bodies, and top-level comments. In parallel, `gemma-3-12b-it` [19] generated concise, context-preserving queries from each post, capturing user intent to support recommendation and retrieval.

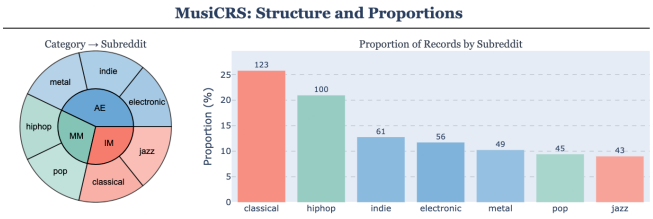

Fig. 3: Dataset genre and entity coverage (left) and dialogue structure statistics (right). MusiCRS encompasses diverse genres, artists, and song distributions, and a wide range of conversation styles. **Macro-genre abbreviations:** MM (Modern & Mainstream: *hip-hop*, *pop*); IM (Instrumental / Art Music: *classical*, *jazz*); AE (Alternative / Experimental: *indie*, *metal*, *electronic*).

Query Normalization and Enhanced Candidate Selection. For evaluation, we constructed candidate pools that include the ground truth entities while remaining realistic. For each conversation, we take the top 10 ground-truth entities (ranked by comment upvotes) and add about 90 additional entities randomly sampled from the same subreddit. We then shuffle the pool to remove positional bias. This design provides broad evaluation coverage while preserving topical coherence within each music community.

Corpus Refinement and Audio Grounding. We refine the dataset by retaining conversations with 2–30 musical entities, focusing on substantive discussions while eliminating noise. We cap each comment at 5 entities to prevent verbosity and topic drift [20]. Ground truth entities are extracted from comments and ranked by upvote score. We use upvotes as a popularity signal while maintaining diverse subreddit representation, balancing mainstream and niche musical content [21]. YouTube links were validated and segmented into audio clips, grounding each conversation in authentic music content for evaluation. Fig. 2 illustrates genre-specific conversations, including the derived query, audio clip, and candidate pools.

Human Validation. To ensure conversational quality and audio grounding, four graduate student annotators manually validated $1,623$ conversations. Each was categorized as *music*, *pop culture*, or *other*, including YouTube link verification and entity relevance review. During this process, 41.3% of URLs were discarded as invalid, while 10.5% were corrected or added through manual verification. Entity refinement removed 20.0% of irrelevant entities and recovered 0.3% of missing entities, improving overall precision. Additionally, 31.1% of conversations were reassigned to the correct categories, ensuring proper alignment with the music-focused benchmark. Spam and off-topic content were removed throughout this validation process. For the final MusiCRS benchmark, we retained only the conversations classified as *music*, resulting in 477 high-quality music-focused conversational threads.

2.2. Dataset Statistics

The curated MusiCRS benchmark contains 477 conversational threads spanning $3,589$ musical entities, with an average of 8 ground truth entities per thread. Each thread includes 100 evaluation candidates, constructed by combining the ground-truth items with subreddit-specific distractors to form realistic recommendation lists. Fig. 3 summarizes genre balance and subreddit proportions: Classical (25.8%), Hip-Hop (21.0%), Indie (12.8%), Electronic (11.7%), Metal (10.3%), Pop (9.4%), and Jazz (9.0%). This distribution ensures comprehensive coverage across mainstream and niche musical content, capturing variation in conversational styles, knowledge